Differentiation of Trigonometric Functions from First Principles

Exam code: 9MA0

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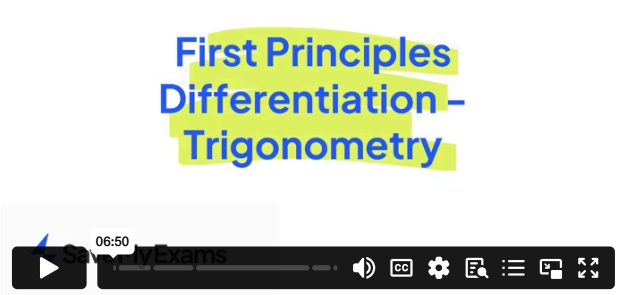

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Differentiation of trigonometric functions from first principles

How do I find derivatives of trigonometric functions from first principles?

- Recall that for a function **f(x)** the definition of the **derivative from first principles** (see First Principles Differentiation) is:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

- The derivatives of the trigonometric functions depend on the following **small angle approximations**
  - When **θ** is small (i.e. close to zero) and measured in radians then

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$$\tan\theta \approx \theta$$

- The small angle approximations allow us to produce the following intermediate limit results:

$$\lim_{h\to 0}\frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h\to 0}\frac{\cos h - 1}{h} = 0$$

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i.e. $\lim_{h\to 0}\frac{\sin h}{h} = \lim_{h\to 0}\frac{h}{h} = \lim_{h\to 0} 1 = 1$

$\sin h \approx h$ WHEN h IS CLOSE TO ZERO

$\lim_{h\to 0}\frac{\cos h - 1}{h} = \lim_{h\to 0}\frac{1-\frac{1}{2}h^2-1}{h} = \lim_{h\to 0}\left(-\frac{1}{2}h\right) = 0$

$\cos h \approx 1-\frac{1}{2}h^2$ WHEN h IS CLOSE TO ZERO

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- And those intermediate results allow us to find the derivatives of **sin** and **cos**

1. IF $y = \sin x$ THEN $\frac{dy}{dx} = \cos x$
2. IF $y = \cos x$ THEN $\frac{dy}{dx} = -\sin x$

DON'T FORGET THE MINUS SIGN WITH THE DERIVATIVE OF cos!

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Help me understand

$$f'(x) = \lim_{h \to 0} \frac{\quad}{h}$$

$\sin(A+B) = \sin A\cos B + \cos A\sin B$

$$= \lim_{h \to 0} \frac{\sin x \cosh + \cos x \sinh - \sin x}{h}$$

$$= \lim_{h \to 0} \left(\sin x \left(\frac{\cosh - 1}{h} \right) + \cos x \left(\frac{\sinh}{h} \right) \right)$$

$$= \sin x(0) + \cos x(1) = \cos x$$

$\lim_{h \to 0} \frac{\cosh - 1}{h} = 0$ $\qquad$ $\lim_{h \to 0} \frac{\sinh}{h} = 1$

SO THE DERIVATIVE OF $\sin x$ IS $\cos x$

IF $f(x) = \cos x$ THEN

$$f'(x) = \lim_{h \to 0} \frac{\cos(x+h) - \cos x}{h}$$

$\cos(A+B) = \cos A\cos B - \sin A\sin B$

$$= \lim_{h \to 0} \frac{\cos x \cosh - \sin x \sinh - \cos x}{h}$$

$$= \lim_{h \to 0} \left(\cos x \left(\frac{\cosh - 1}{h} \right) - \sin x \left(\frac{\sinh}{h} \right) \right)$$

$$= \cos x(0) - \sin x(1) = -\sin x$$

SO THE DERIVATIVE OF $\cos x$ IS $-\sin x$

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How do I find derivatives of other trigonometric functions?

- The derivative of **tan** is given by the following formula:

3. IF $y = \tan x$ THEN $\frac{dy}{dx} = \sec^2 x$

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- The easiest way to derive this is to use the **quotient rule** and the derivatives of **sin** and **cos**
- But it can also be derived from first principles using the small angle approximation for **tan** (see the Worked Example)
- The general formulae for the derivatives of the trigonometric functions are:

4. IF $y = \sin kx$ THEN $\frac{dy}{dx} = k\cos kx$

5. IF $y = \cos kx$ THEN $\frac{dy}{dx} = -k\sin kx$

6. IF $y = \tan kx$ THEN $\frac{dy}{dx} = k\sec^2 kx$

DON'T FORGET THE MINUS SIGN WITH THE DERIVATIVE OF cos!

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- These formulae follow from combining the derivatives of the three basic functions with the **chain rule**, but they are worth knowing on their own

- Remember that when doing calculus with trigonometric functions you have to measure angles in **radians.**
- The formula for the derivative of **tan *x*** is included in the exam formulae booklet.
- The derivatives of **sin *x*** and **cos *x*** are NOT included in the formula booklet – you have to know them.
- The small angle approximations for **cos *x***, **sin *x*** and **tan *x*** are included in the exam formulae booklet – you don't have to memorise them.
- Be sure to read first principle differentiation exam questions clearly – they will state any results you can treat as 'givens' in your answer.

Worked Example

Teacher only

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